

Reinforcement Learning.

policy optimization. Key Problem

1. since gradient **varies sharply**, make it difficult converging.

each steps occur **correlation in trajectories**.

$\pi_\theta(a|s)$ is the optimization objectives.

$$\therefore D(\theta) = E_{a \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t \cdot R_t \right] = E_{a \sim \pi_\theta} [R(t, a)]$$

$$= \int R(t, a) \cdot \pi_\theta(a) da$$

$$\text{by } \nabla_\theta \pi_\theta(a) = \pi_\theta(a) \cdot \nabla_\theta \ln \pi_\theta(a) \left[d \ln x = \frac{dx}{x} \right].$$

$$\Rightarrow \nabla_\theta D = \int R(t, a) \cdot \nabla_\theta \pi_\theta(a) da$$

$$= \int R(t, a) \cdot \pi_\theta(a) \cdot \nabla_\theta \ln \pi_\theta(a) da.$$

$$= E_{a \sim \pi_\theta} [R(t, a) \cdot \nabla_\theta \ln \pi_\theta(a)].$$

Take advantage $A = R(t, a) - B(s)$, $B(s)$ is baseline in J .

$$\text{by } \int B(s) \cdot \nabla_\theta \pi_\theta(a) da = B(s) \cdot \nabla_\theta \int \pi_\theta(a) da$$

$$= B(s) \cdot \nabla_\theta 1 = 0.$$

$$\Rightarrow E_{a \sim \pi_\theta} [R(t, a) \cdot \nabla_\theta \ln \pi_\theta(a)] = E_{a \sim \pi_\theta} [A \cdot \nabla_\theta \ln \pi_\theta(a)]$$

$$\therefore \text{A gradient at time } t \text{ is } g_t = \nabla_\theta \log \pi_\theta(a_t | s_t) \cdot A_t.$$

but $\forall i > j$, g_i, g_j shares the same trajectories.

$$\therefore \text{cov}(g_i, g_j) > 0.$$

$$\text{Take } \bar{g} = \frac{1}{T} \sum_{t=1}^T g_t \Rightarrow \text{var}(\bar{g}) = \text{var}\left(\frac{1}{T} \sum_{t=1}^T g_t\right)$$

$$\text{by } \text{var}(A+B) = \text{var}(A) + \text{var}(B) + 2 \cdot \text{cov}(A, B).$$

$$\Rightarrow \text{var}\left(\frac{1}{T} \sum_{t=1}^T g_t\right) = \frac{1}{T^2} \sum_{i=1}^T \sum_{j=1}^T \text{cov}(g_i, g_j). \text{ not eliminate.}$$

If $\text{cov}(f_i, f_j) = 0$, then $\text{Var}(\bar{f}) = \frac{1}{T}$. $T \cdot \text{Var}(\bar{f}) = \frac{1}{T} \cdot \text{Var}(f)$.

here see f as homogeneous

Here. Replacing $R(t, a)$ with A is a control variate.

The advantage can be written as. $Q(\pi, a) - V(\pi)$.